Name: Harsh Kawinna

Class: D15A

Roll No: 60

Experiment 1 – Design Star and Snowflake Schema

Aim:

To design and understand Star Schema and Snowflake Schema for a Banking and Transaction Data Warehouse, using transaction data as the fact and Customer, Account, Branch, Employee, Date, and Transaction Type as dimensions.

Objective

To understand the concept of **Data Warehouse Schema**.

To design a **Star Schema** for banking transaction data.

To design a **Snowflake Schema** for banking transaction data.

To understand the difference between Star Schema and Snowflake Schema.

Theory:

A Data Warehouse is a centralized system used to store and analyze large amounts of historical data for decision-making and reporting. In banking systems, huge volumes of transactions are generated every day, making an efficient data warehouse design important for analysis. Star Schema and Snowflake Schema are two commonly used dimensional models in data warehousing. In a Star Schema, a central Fact Table is directly connected to multiple Dimension Tables. The dimensions are generally denormalized, making the schema simple and queries faster. In a Snowflake Schema, the dimension tables are further normalized into sub-dimension tables. This reduces data redundancy but increases the number of tables and joins. For this banking system, FACT_TRANSACTION stores measurable transaction information such as transaction amount, transaction fee, and balance, while dimensions provide descriptive information about customers, accounts, branches, employees, dates, and transaction types

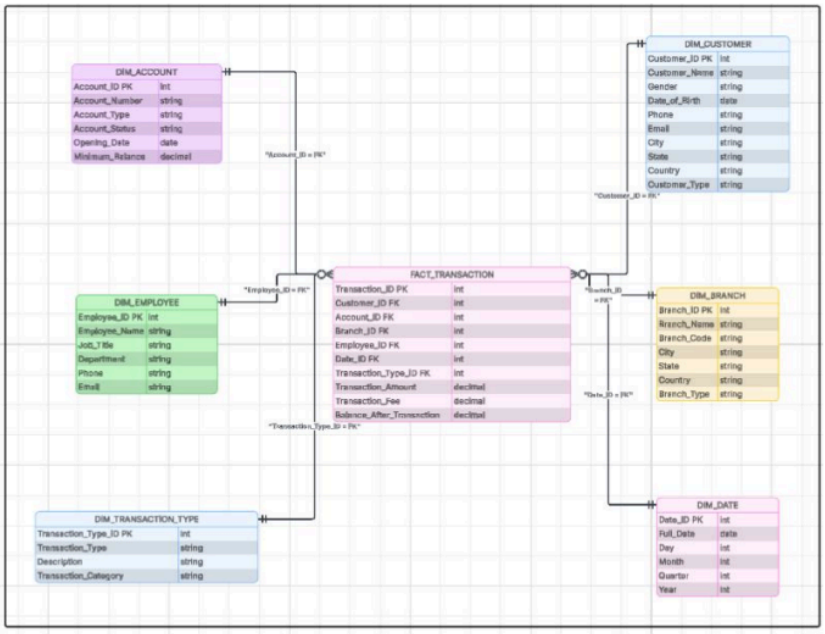
1. Star Schema:

This is a Star Schema designed for banking transaction data. The central table is FACT_TRANSACTION, which stores transaction measures such as transaction amount, transaction fee, and balance after transaction. It is connected directly to

six dimension tables: Customer, Account, Branch, Employee, Transaction Type, and Date. The dimension tables provide descriptive information about each transaction. Primary keys in the dimension tables become foreign keys in the fact table. Since all dimension tables are directly connected to the central fact table, the structure resembles a star, hence it is called a Star Schema

Structure:

Star Schema:



---

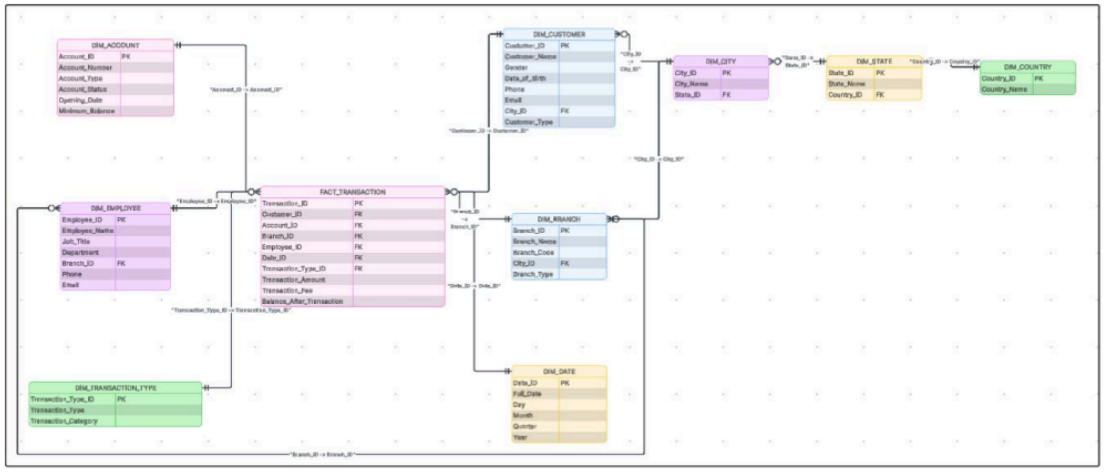
2. Snowflake Schema:

This is a Snowflake Schema designed for banking transaction data. The central table is FACT_TRANSACTION, which stores transaction details such as transaction amount,

transaction fee, and balance after transaction. It is connected to dimension tables such as Customer, Account, Branch, Employee, Transaction Type, and Date. In this schema, some dimensions are further normalized. For example, the Customer dimension is connected to City, City is connected to State, and State is connected to Country. Similarly, the Branch dimension is connected to the geographical hierarchy. This normalization reduces data redundancy and creates a hierarchical structure, which gives the schema its snowflake shape.

Structure:

Snowflake Schema:



Result:

The **Star Schema** and **Snowflake Schema** for the **Banking Transaction System** were successfully designed.